

Mimic fasting. *Without fasting.*

A molecule made by your gut bacteria activates the same longevity pathways as **caloric restriction**.

11 SLIDES
→ SWIPE

• ACLITHO

lithocholic acid · noun

A *secondary* bile acid, produced by your gut microbiome from the bile salts your liver secretes.

CELLULAR SIGNATURE

AMPK ↑
ENERGY THERMOSTAT

NAD⁺ ↑
SIRTUIN COFACTOR

autophagy ↑
CELLULAR CLEANUP

Every known longevity pathway converges on a single biological target.

AMPK

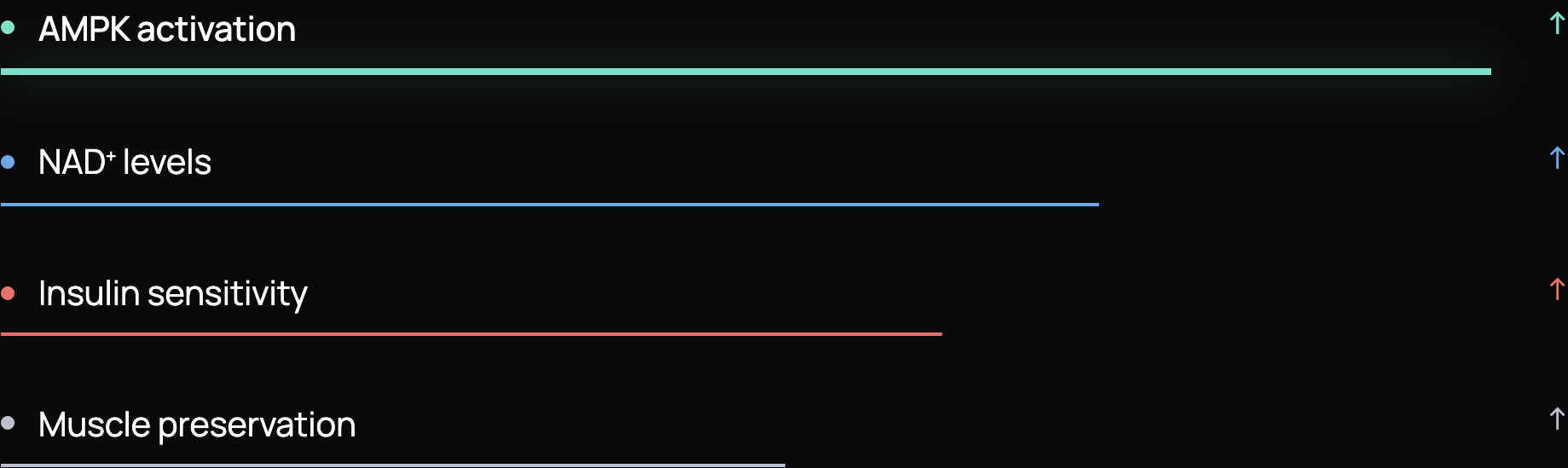
AMP-activated protein kinase

The cellular energy thermostat. It senses low ATP — fasting, exercise — and triggers the cascade: [mitochondrial biogenesis](#), [autophagy](#), [brake on mTORC1](#).

Your bacteria manufacture a *fasting mimetic*.

01	Liver	Synthesises primary bile acids from cholesterol.
02	Small intestine	Digests fats. 95 % is reabsorbed.
03	Colon	5 % reaches the bacteria.
04	Microbiome	Lactobacillus, Clostridium, Eubacterium. 7 α -dehydroxylase enzyme.
05	ACLITHO	Secondary bile acid. Crosses the barrier. Acts as a distant hormone.

The same adaptations as prolonged fasting. Without the fast.



Work published in Nature · direct AMPK activation observed across multiple tissues, including muscle and brain.

A simple question: what if a molecule could activate AMPK *without* requiring you to fast?



Intestinal bile acids act as signaling hormones on host metabolism through the gut–liver–microbiome axis.

– Wahlström et al., Cell Metabolism
2016 · Karolinska Institute

How to **raise** your ACLITHO naturally.

	LEVER	MECHANISM	IN PRACTICE
01	● Intermittent fasting	Reduces enterohepatic recycling. More bile acids reach the bacteria.	16/8 WINDOW · 3-5×/WEEK
02	● Cyclical ketosis	Shifts synthesis toward ACLITHO-producing pathways.	5-7 DAY CYCLES · 1-2×/MONTH
03	● Physical exercise	Increases microbiome diversity and enterohepatic circulation.	150 MIN/WK MODERATE · 2× STRENGTH
04	● Liver health	Guarantees primary bile acid production upstream.	LIMIT ALCOHOL · CONTROL FRUCTOSE

Three are free. The fourth is mostly about avoiding things, not consuming them.

Without a healthy liver, *nothing* downstream works.

30%

of adults have **silent hepatic steatosis**. Less bile acid, less ACLITHO, less AMPK: the loop closes.

steatosis → ↓ bile acids → ↓ ACLITHO → ↓ AMPK → insulin resistance

Here's what I prescribe in clinic.



Install intermittent fasting 16/8 window · 3-5×/week

The most powerful and simplest lever. If you're starting out, begin with 14/10.



Add two strength sessions 30-45 min · large muscle groups

Resistance exercise improves microbiome diversity and enterohepatic circulation. Bodyweight is enough.



Cut added fructose 30 days · sodas, syrups, ultra-processed

The main fuel for hepatic steatosis, which blocks the whole chain upstream. Whole fruit is still fine.



Check your liver markers ALT · AST · GGT

Silent steatosis is more common than you think. If elevated, talk to your doctor.

Longevity **without** deprivation.

01

AMPK is the common target of every known longevity pathway. It's the master switch.

02

ACLITHO, produced by your bacteria, activates AMPK the way prolonged fasting would.

03

Four concrete levers, no supplement or medical protocol: fasting, ketosis, exercise, liver.

04

Without a healthy liver, the whole chain collapses. Check your liver markers.



→ COMMENT

Comment "ACLITHO"

Type "ACLITHO" in the comments and I'll send you the full article — the detailed biochemistry, the 4-week protocol, and the liver markers to ask your doctor about.

DR. GUÉNOLÉ ADDOR — FMH

Every day counts.

SOURCES

Cell Metabolism 2016
NEJM 2019 · Nature Cell Bio 2018